

3-1-1: Examining Switching and Routing Issue

At the end of this episode, I will be able to:

1. Identify and explain the significance of switching and routing issues, given a scenario.

Learner Objective: *Identify and explain the significance of switching and routing issues, given a scenario*

Description: In this episode, the learner will examine switching and routing issues that can affect network traffic. We will examine network loops, port roles, root bridge election, route selection issues and more.

- What Layer 2 services, such as STP, can cause network issues?
 - o Forwarding loops
 - ♣ Symptom
 - ♣ Unintended traffic loops causing broadcast storms and network slowdowns
 - ♣ Common causes
 - ♣ Inadequate STP configuration
 - ♣ Failure to block redundant paths
 - ♣ Solution
 - ♣ Verify STP operation to ensure, correct identification and redundant path blocking
 - o Root bridge selection
 - ♣ Symptom
 - ♣ Poor network performance due
 - ♣ Common causes
 - ♣ Inefficient root bridge location
 - ♣ Default configuration leading to an unplanned device becoming the root bridge
 - ♣ Solution
 - ♣ Manually set bridge priority on the preferred root bridge (controlled root bridge election)
 - o Port roles
 - ♣ Root
 - ♣ Non-root switches, having the best cost path to root bridge. These ports forward data to the root bridge.
 - ♣ Designated
 - ♣ Ports on root and designated switches
 - ♣ Blocked
 - ♣ Blocked
 - ♣ All other ports to bridges or switches are in a blocked state.
 - ♣ Symptom
 - ♣ Misrouted traffic or network segments becoming isolated
 - ♣ Common causes
 - ♣ Incorrect STP calculations or configuration errors assigning incorrect port roles
 - ♣ Solution
 - ♣ Check STP port roles (root, designated, blocked)
 - ♣ Adjust configurations as needed
 - o Port states
 - ♣ States
 - ♣ Disabled - The port is disabled, does not forward traffic
 - ♣ Blocking - In a blocking state, does not forward traffic
 - ♣

- - Listening - listens for and sends BPDUs
 - ♣ Learning - When a superior BPDU is received, it will stop sending its own BPDUs, and will start relaying the superior BPDUs.
 - ♣ Forwarding - Forwarding traffic.
- ♣ Symptom
 - ♣ Slow network convergence
 - ♣ Devices unable to communicate immediately after network changes
- ♣ Common causes
 - ♣ Ports stuck in inappropriate states (listening, learning) for too long due to STP settings
- ♣ Solution
 - ♣ Review and adjust STP timers and settings
 - ♣ Ensure ports transition to the forwarding state in a timely manner
- Configuration issues on managed switches, could this be the cause?
 - o Incorrect VLAN Assignment
 - ♣ Symptom
 - ♣ Lack of connectivity
 - ♣ Access issues (including unauthorized)
 - ♣ Common causes
 - ♣ Assigning a device or port to the wrong VLAN
 - ♣ Solution
 - ♣ Ensure port VLAN configurations match the intended network segment for the connected devices
 - o ACLs (Access Control Lists)
 - ♣ Used to permit or deny traffic through the network, impacting access and flow
 - ♣ Symptom
 - ♣ Blocking legitimate traffic
 - ♣ Allowing unauthorized
 - ♣ Common cause
 - ♣ Incorrectly configured ACLs
 - ♣ Solution
 - ♣ Regularly review and update ACLs
 - ♣ align with network security policies and access requirements
- How about Layer 3 or routing issues?
 - o Routing tables
 - ♣ Symptom
 - ♣ Misdirected or dropped traffic, inability to reach specific network segments.
 - ♣ Common causes
 - ♣ Incorrectly configured static routes, outdated or missing routes due to dynamic routing failures.
 - ♣ Solution
 - ♣ Verify the accuracy of routing table entries, update static routes, and ensure dynamic routing protocols are properly configured and operational.
 - o Default routes
 - ♣ Symptom
 - ♣ Inability to access external networks or the internet.
 - ♣ Common causes
 - ♣ Absence of a default route or incorrect default gateway configuration.
 - ♣ Solution
 - ♣ Ensure that a correct default route is set and points to a valid gateway that can route traffic outside the local network.

Additional Resources:

- o If applicable